

CHAPTER 6

GRAPHS OF TRIGONOMETRIC FUNCTIONS

6.1 INTRODUCTION

We have already studied in the previous chapters that all trigonometric functions are periodic. For example, sine and cosine functions are periodic with period 2π , while tangent and cotangent functions are periodic with period π . We know that if $f(x)$ is a periodic function with period T and $a > 0$, then $f(ax + b)$ is periodic with period T/a . Therefore, $\sin(ax + b)$ and $\cos ax$ are periodic functions with period $\frac{2\pi}{a}$. If the graph of a periodic function with period T is to be

drawn in a given interval, then it is sufficient to draw its graph only in an interval of length T . Because, once it is drawn in one such interval, it can be easily drawn completely by repeating it over the intervals of lengths T . The amplitude of a function is defined as the greatest numerical value which it can attain.

Using the knowledge acquired in the above discussion let us now draw the graphs of various trigonometric functions.

6.2 GRAPHS OF TRIGONOMETRIC FUNCTIONS

6.2.1 GRAPH OF $y = \sin x$

Since $\sin x$ is a periodic function with period 2π . So, we will draw the graph of $y = \sin x$ in the interval $[0, 2\pi]$. In order to draw the graph of $y = \sin x$ in the interval $[0, 2\pi]$, we first draw it in $[0, \pi/2]$. The values of $\sin x$ for different values of x in the interval $[0, \pi/2]$ are given in the following table.

x	0°	30°	45°	60°	90°
$\sin x$	0	$1/2 = 0.5$	$1/\sqrt{2} = 0.707$	$\sqrt{3}/2 = 0.866$	1

Using the above table and the fact that $\sin x$ is an increasing function we obtain the graph of $y = \sin x$ in the interval $[0, \pi/2]$ as shown in Fig. 6.1. In the interval $[\pi/2, \pi]$, we draw the graph of $y = \sin x$ by using the fact that $\sin(\pi - x) = \sin x$. Finally, we draw it in the interval $[\pi, 2\pi]$, using the fact that $\sin(\pi + x) = -\sin x$ which means that the graph of $y = \sin x$ in $[\pi, 2\pi]$ is the mirror image of the graph of $y = \sin x$ in $[0, \pi]$. The graph is now sketched in Fig. 6.1.

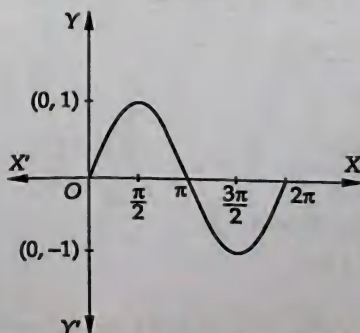


Fig. 6.1 Graph of $y = \sin x$

Since $\sin x$ is a periodic function with period 2π . Therefore, we use the following algorithm to draw the graphs of $y = c \sin ax$.

ALGORITHM

STEP I Obtain the values of a and c .

STEP II Draw the graph of $y = \sin x$ and mark the points where it crosses x -axis.

STEP III Divide the x -coordinates of the points where $y = \sin x$ crosses x -axis by a and mark maximum and minimum values of $y = c \sin ax$ as c and $-c$ on y -axis.

The graph so obtained is the graph of $y = c \sin ax$ as shown in Fig. 6.2.

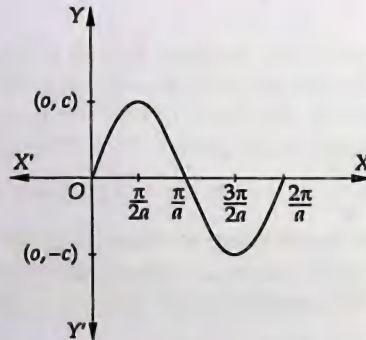


Fig. 6.2 Graph of $y = c \sin ax$

EXAMPLE 1 Sketch the graph of $y = 3 \sin 2x$.

SOLUTION To obtain the graph of $y = 3 \sin 2x$ we first draw the graph of $y = \sin x$ in the interval $[0, 2\pi]$ and then divide the x -coordinates of the points where it crosses x -axis by 2. The maximum and minimum values are 3 and -3 respectively as shown in Fig. 6.3.

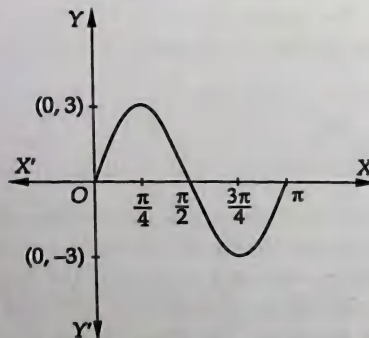


Fig. 6.3 Graph of $y = 3 \sin 2x$

EXAMPLE 2 Sketch the graph of $y = 3 \sin (2x - 1)$.

SOLUTION We have,

$$y = 3 \sin (2x - 1)$$

$$\Rightarrow (y - 0) = 3 \sin 2 \left(x - \frac{1}{2} \right) \quad \dots(i)$$

Shifting the origin at $(1/2, 0)$, we obtain

$$x = X + \frac{1}{2} \text{ and } y = Y + 0$$

Substituting these values in (i), we get

$$Y = 3 \sin 2X$$

Thus, if we draw the graph of $Y = 3 \sin 2X$ and shift it by $1/2$ unit to the right, we get the required graph as shown in Fig. 6.4.

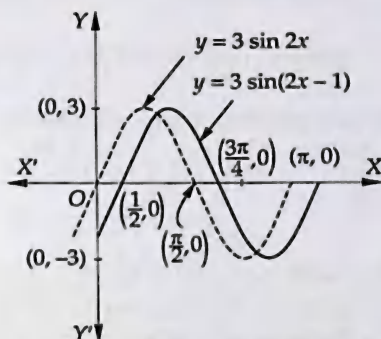


Fig. 6.4 Graph of $y = 3 \sin(2x - 1)$

EXAMPLE 3 Sketch the graph of $y = \sin x$ and $y = \sin 2x$ on the same axes.

SOLUTION Clearly $\sin 2x$ is a periodic function with period $2\pi/2 = \pi$ whereas $\sin x$ is periodic with period 2π . The graphs of $y = \sin x$ and $y = \sin 2x$ on different axes are shown in Figs. 6.5 and 6.6 respectively.

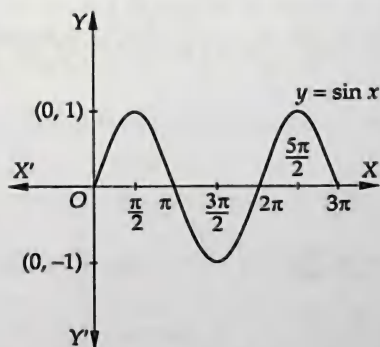


Fig. 6.5 Graph of $y = \sin x$

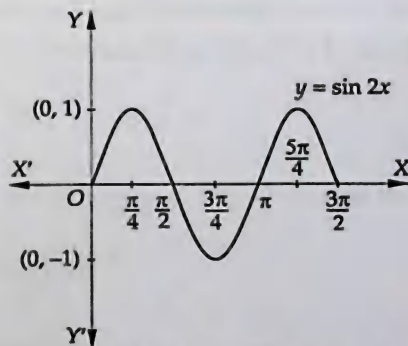


Fig. 6.6 Graph of $y = \sin 2x$

If these two graphs are drawn on the same axes, then the graphs are drawn in Fig. 6.7.

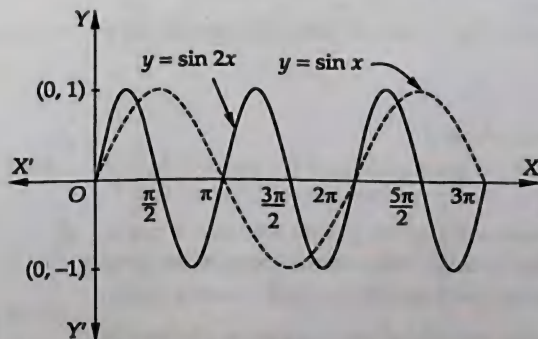


Fig. 6.7 Graphs of $y = \sin x$ and $y = \sin 2x$ on the same scale

EXERCISE 6.1

1. Sketch the following graphs:

(i) $y = 2 \sin 2x$

(ii) $y = 3 \sin x$

(iii) $y = 2 \sin \left(x - \frac{\pi}{4} \right)$

(iv) $y = 2 \sin (2x - 1)$

(v) $y = 3 \sin (3x + 1)$

(vi) $y = 3 \sin \left(2x - \frac{\pi}{4} \right)$

2. Sketch the graph of the following pairs of functions on the same axes :

(i) $y = \sin x, y = \sin \left(x + \frac{\pi}{4} \right)$

(ii) $y = \sin x, y = \sin 3x$

6.2.2 GRAPH OF $y = \cos x$

Since $\cos x$ is a periodic function with period 2π . So, it is sufficient to draw the graph in the interval $[0, 2\pi]$. We first draw the graph in the interval $[0, \pi/2]$. For this we use the table given below and the fact that $\cos x$ is a decreasing function in this interval.

x	0°	30°	45°	60°	90°
$\cos x$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0

Now, to draw the graph of $y = \cos x$ in the interval $[\pi/2, \pi]$ we use the relation $\cos(\pi - x) = -\cos x$. If we give values of x between 0 and $\pi/2$, then this relation shows that the values of $\cos x$ between $\pi/2$ and π are negative of its values in the interval $[0, \pi/2]$. Thus, the graph of $y = \cos x$ between $\pi/2$ and π is below x -axis. Now, $\cos(\pi + x) = -\cos x$ shows that the graph of $y = \cos x$ between π and 2π is the mirror image in x -axis of its graph between 0 and π .

Making use of all these results we obtain the sketch of the graph of $y = \cos x$ as shown in Fig. 6.8.

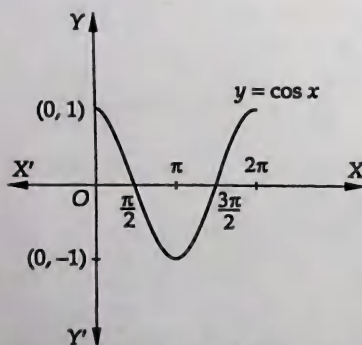


Fig. 6.8 Graph of $y = \cos x$

In order to draw the graph of $y = c \cos ax$ from the graph of $y = \cos x$, we may use the following algorithm.

ALGORITHM

STEP I Obtain the values of a and c .

STEP II Draw the graph of $y = \cos x$ and mark the points where it crosses x -axis.

STEP III Divide the x -coordinates of the points where $y = \cos x$ meets x -axis by a and also mark maximum and minimum values of $y = c \cos ax$ as c and $-c$ on y -axis.

The graph so obtained is the graph of $y = c \cos ax$ as shown in Fig. 6.9.

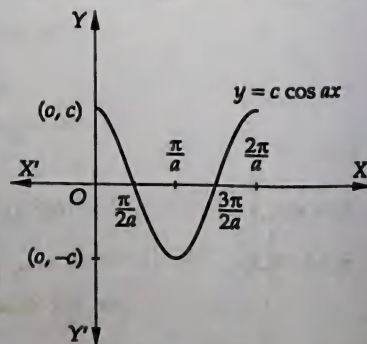


Fig. 6.9 Graph of $y = c \cos ax$

ILLUSTRATIVE EXAMPLES

EXAMPLE 1 Draw the sketch of the graph of $y = 3 \cos 2x$.

SOLUTION Replacing c by 3 and a by 2 in the graph of $y = c \cos ax$, we obtain the graph of $y = 3 \cos 2x$ as shown in Fig. 6.10.

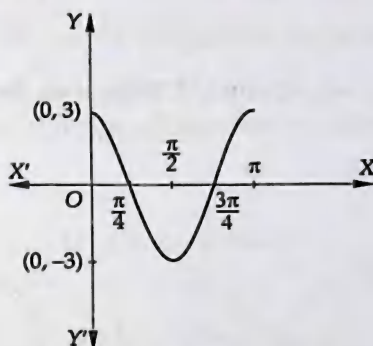


Fig. 6.10 Graph of $y = 3 \cos 2x$

EXAMPLE 2 Sketch the graph of $y = \cos\left(x - \frac{\pi}{4}\right)$.

SOLUTION We have,

$$y = \cos\left(x - \frac{\pi}{4}\right) \Rightarrow y - 0 = \cos\left(x - \frac{\pi}{4}\right) \quad \dots(i)$$

Shifting the origin at $(\pi/4, 0)$, we obtain

$$x = X + \frac{\pi}{4}, \quad y = Y + 0$$

Substituting these values in (i), we get

$$Y = \cos X.$$

Thus, to draw the graph of $y = \cos\left(x - \frac{\pi}{4}\right)$ we first draw the graph of $y = \cos x$ and then shift it through $\pi/4$ units to the right. The graph is drawn in Fig. 6.11.

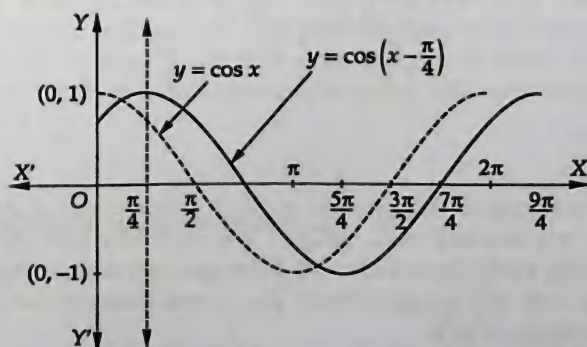


Fig. 6.11 Graph of $y = \cos\left(x - \frac{\pi}{4}\right)$

EXAMPLE 3 Sketch the graphs of $y = \cos 2x$ and $y = \cos\left(2x - \frac{\pi}{4}\right)$ on the same scale.

SOLUTION We have,

$$y = \cos\left(2x - \frac{\pi}{4}\right) = \cos 2\left(x - \frac{\pi}{8}\right) \Rightarrow y - 0 = \cos 2\left(x - \frac{\pi}{8}\right)$$

Shifting the origin at $(\pi/8, 0)$, we have

$$x = X + \frac{\pi}{8} \text{ and } y = Y + 0$$

Using these relations $y - 0 = \cos 2\left(x - \frac{\pi}{8}\right)$ reduces to $Y = \cos 2X$.

It follows from the above discussion that the graph of $y = \cos\left(2x - \frac{\pi}{4}\right)$ is similar to the graph of $y = \cos 2x$ but it lags the graph of $y = \cos 2x$ by $\pi/8$. Thus, if we draw the graph of $y = \cos 2x$ and shift it through $\pi/8$ units to the right, we obtain the graph of $y = \cos(2x - \pi/4)$ as shown in Fig. 6.12.

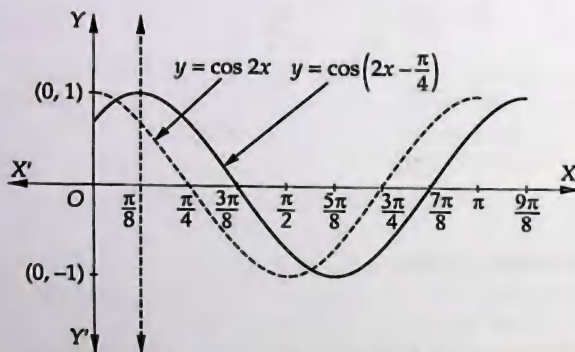


Fig. 6.12 Graph of $y = \cos 2x$ and $y = \cos\left(2x - \frac{\pi}{4}\right)$ on the same scale

EXERCISE 6.2

1. Sketch the following graphs:

(i) $y = \cos\left(x + \frac{\pi}{4}\right)$ (ii) $y = \cos\left(x - \frac{\pi}{4}\right)$ (iii) $y = 3 \cos(2x - 1)$ (iv) $y = 2 \cos\left(x - \frac{\pi}{2}\right)$

2. Sketch the graphs of the following functions on the same scale.

(i) $y = \cos x$ and $y = \cos\left(x - \frac{\pi}{4}\right)$ (ii) $y = \cos 2x$ and $y = \cos 2\left(x - \frac{\pi}{4}\right)$

(iii) $y = \cos x$ and $y = \cos\left(\frac{x}{2}\right)$

6.2.3 GRAPH OF $y = \tan x$

Since $\tan x$ is a periodic function with period π . So, it is sufficient to draw the graph over an interval of length π , in particular $[-\pi/2, \pi/2]$. First we will draw the graph in the interval $[0, \pi/2]$. For this we use the table given below and the fact that $\tan x$ is increasing in this interval. Also, as $x \rightarrow \pi/2$, $\tan x \rightarrow \infty$. So, the graph gets closer and closer to the line $x = \pi/2$ as $x \rightarrow \pi/2$. But it never touches the line $x = \pi/2$.

x	0°	30°	45°	60°	90°
$\tan x$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	∞

Since $\tan(-x) = -\tan x$, therefore, if $(x, \tan x)$ is any point on the curve $y = \tan x$, then $(-x, -\tan x)$ will also be a point on it. This means that the graph is symmetric in opposite quadrants.

Using all the above points we obtain the sketch of the curve $y = \tan x$ as given in Fig. 6.13.

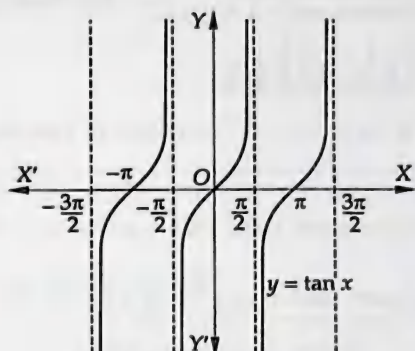


Fig. 6.13 Graph of $y = \tan x$

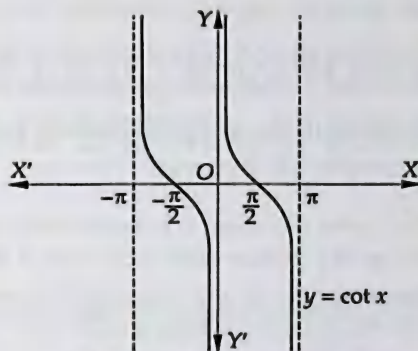


Fig. 6.14 Graph of $y = \cot x$

Proceeding as above we obtain the graphs of other trigonometric functions as shown in Figures 6.14, 6.15, and 6.16.

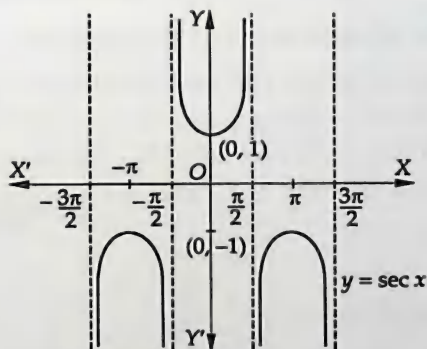


Fig. 6.15 Graph of $y = \sec x$

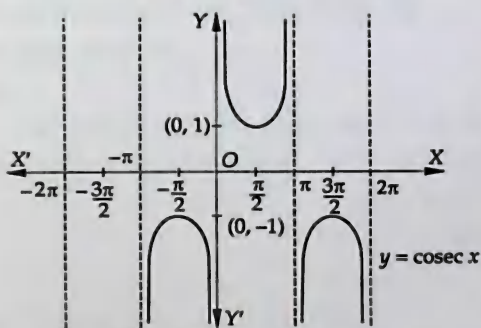


Fig. 6.16 Graph of $y = \operatorname{cosec} x$

It should be noted that $\operatorname{cosec} x$ and $\sec x$ are also periodic functions with period 2π while $\cot x$ is periodic with period π . As $x \rightarrow \pm \pi/2$, $\sec x \rightarrow \infty$. So, the curve comes closer and closer to ∞ as $x \rightarrow \pm \pi/2$. These lines are known as the asymptotes to the curve. Similarly, $x = 0, x = \pi$ etc. are asymptotes of $y = \operatorname{cosec} x$.

EXERCISE 6.3

Sketch the graphs of the following functions:

1. $y = \sin^2 x$

2. $y = \cos^2 x$

3. $y = \sin^2 \left(x - \frac{\pi}{4} \right)$

4. $y = \tan 2x$

5. $y = 2 \tan 3x$

6. $y = 2 \cot 2x$

Sketch the graphs of the following functions on the same scale:

7. $y = \cos 2x, y = \cos \left(2x - \frac{\pi}{3} \right)$

8. $y = \sin^2 x, y = \sin x$

9. $y = \tan x, y = \tan^2 x$

10. $y = \tan 2x, y = \tan x$

SUMMARY

1. The curve $y = \sin x$ is symmetric in opposite quadrants and $-1 \leq y \leq 1$.
2. The curve $y = \cos x$ is symmetric about y -axis and $-1 \leq y \leq 1$.
3. The curve $y = \tan x$ is symmetric in opposite quadrants and $-\infty < y < \infty$.
4. The curve $y = \sec x$ is symmetric about y -axis and $y \geq 1$ or $y \leq -1$. The values of y do not exist for $x = (2n + 1) \frac{\pi}{2}, n \in \mathbb{Z}$.
5. The curve $y = \operatorname{cosec} x$ is symmetric in opposite quadrants such that $y \geq 1$ or $y \leq -1$. The values of y do not exist for $x = n\pi, n \in \mathbb{Z}$.
6. The curve $y = \cot x$ is symmetric in opposite quadrants such that $-\infty < y < \infty$. The values of y do not exist for $x = n\pi, n \in \mathbb{Z}$.